

Problem 9

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Problem 1 Suppose you own a tour bus and you book groups of 20 to 80 people for a day tour. The cost per person is \$30 minus \$0.25 for every ticket sold. If gas and other miscellaneous costs are \$300, how many tickets should you sell to maximize your profit? Treat the number of tickets as a nonnegative real number.

Explanation. Let x denote the number of tickets sold. Then the revenue, R is given by

$$R(x) = x(30 - x \cdot 0.25).$$

Therefore, the profit, P , is given by

$$P(x) = R(x) - C(x) = x(30 - x \cdot 0.25) - 300.$$

So, we have to find the maximum of the function P on its domain, $[20, 80]$. Since P is continuous on the closed interval, the Extreme Value Theorem guarantees that the maximum of P is attained on $[20, 80]$. We have to find the critical points of P and evaluate the function at critical points and at end points, and compare the values.

Since

$$P'(x) = 30 - 0.5x$$

we have to solve the equation

$$30 - 0.5x = 0.$$

It follows that the function P has its only critical points at $x = 60$. Since

$$P(60) = 60(30 - 60 \cdot 0.25) - 300 = 60 \cdot 15 - 300 = 600,$$

$$P(20) = 20(30 - 20 \cdot 0.25) - 300 = 20 \cdot 25 - 300 = 200,$$

$$P(80) = 80(30 - 80 \cdot 0.25) - 300 = 800 - 300 = 500,$$

it follows that the maximum profit is \$600 and it is attained when 60 tickets are sold, i.e. at $x = 60$.